

What the Higgs Potential Cannot See:

bulk matter that cannot help select a boundary condition

Winding parity is the $O(4)$ component label, so the potential is one operator traced twice — a dimension and an index; the boundary-condition sign rides on the index alone, and goes blind on a class we classify and count in closed form

Part V — 6D $SU(4)$ gauge-Higgs unification: the published one-loop coefficients of three separate papers reproduced from representation theory alone, and a selection rule on the matter content that costs nothing to apply

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Abstract

Parts I–IV of this series built a piece of representation theory and left its physical purpose implicit. This paper supplies it. Part III’s own translation matrix $U_6 = P_2 P_0^{-1}$ has determinant -1 , so it does not lie in the identity component of the structure group; the consequence is that the parity of the winding number is the $O(4)$ component label, and the entire one-loop Wilson-line potential of the model is *two Schur specialisations of one partition*, $\Sigma_\lambda(t) = s_\lambda(1, 1, t, t^{-1})$ on the identity component and $D_\lambda(t) = s_\lambda(1, -1, t, t^{-1})$ — Part IV’s object — on the reflection coset. The claim is anchored to numbers we did not produce: the standard form in which this field writes its one-loop potential, $\{A + B(-1)^{k_2}\}$, has coefficients that *are* $(\Sigma_\lambda \pm D_\lambda)/2$, and the published numbers of three separate papers — in three groups, two dimensions and two orbifolds — are reproduced from representation theory with nothing fitted. The pair is one object read twice: with σ the orbifold parity and g_t the Wilson-line element, σg_t is the coset alphabet, so $\Sigma_\lambda = \text{Tr}(g_t)$ and $D_\lambda = \text{Tr}(\sigma g_t)$ — a trace and a twisted trace of one element on one space, a dimension and an index. What follows is a statement about what such a potential can and cannot see. Each matter multiplet carries a discrete boundary-condition sign — in the notation Haba, Hosotani and Kawamura have used since 2004, the product $\eta = \eta_0 \eta_1$ of the two signs attached to the orbifold parities — and that sign multiplies the coset character and nothing else. Hence: $(\lambda, \eta) \sim (\lambda^*, \eta(-1)^{\lambda_1})$ is an observational identity — proved, and verified over the range swept to be the only collision there is — and η is unobservable *exactly* on $\mathcal{Z}$, Part IV’s vanishing class, whose closed-form classification is therefore a classification of the multiplets that cannot help select a boundary condition. The bridge to the older literature is exact and it widens the statement: for any representation, a pair of orbifold parities sees a multiplet through exactly three characters, $N^{(\varepsilon_0, \varepsilon_1)} = \frac{1}{4}[\dim + \varepsilon_0 \chi(P_0) + \varepsilon_1 \chi(P_1) + \varepsilon_0 \varepsilon_1 \chi(P_0 P_1)]$, and η multiplies the last one alone — so the theorem is not a property of one six-dimensional model but of the $[p; q, r; s]$ classification itself. A corollary settles a difference of regime we expected to have to live with: because Part IV’s closed form has coefficients of one sign, $D_\lambda(1) = 0$ if and only if $D_\lambda \equiv 0$, so the sign is invisible at vanishing Wilson line phase — where the 5D literature computes — if and only if it is invisible at every phase. Three further statements are proved or measured rather than asserted: the blind class is counted in closed form, $\lceil (k+1)^2/2 \rceil$ at $\lambda_1 = 2k+1$ and never for λ_1 even — machine-checked in Lean 4, and chained to Part III’s existing certificate so that the count is a theorem about the bialternant determinant rather than about a predicate chosen to be countable; the asymmetry between the two halves is traced to the frozen letters being a complete set of roots of unity, which

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puts every zero of the coset character on the unit circle and none of the identity character's there, so that the coset amplitude vanishes only at *rational* Wilson line phases; and the same test, applied to the boundary conditions themselves, shows that $\lfloor (N+1)^2/2 \rfloor$ of the $(N+1)^2$ equivalence classes admit a coset sector at all — also machine-checked, and for every rank rather than over a swept range, because the block pair $(\#\{P_0=+1\}, \#\{P_1=-1\})$ turns out to be a complete class invariant — which splits the invisibility question into one geometric case and one representation-theoretic case, only the second of which is ours. One step past a single multiplet is then free: if the non-blind multiplets of a matter content share one effective sign, the content is blind only if every multiplet in it is — so *collective blindness requires sign frustration*, and the degeneracy left for the sequel is the frustrated half of it. Scope is stated and not buried: every statement here is about a *model*, the theorem is *per multiplet*, and everything that runs the machinery forwards — predicting a spectrum rather than excluding one — is deliberately held over to Part VI. *Retracted before publication, and recorded because the record is worth as much as the result*: an earlier draft of §6 claimed the coset half is the only observable in the model that sees charge conjugation. It is not; charge conjugation is invisible here unconditionally, and Kawamura, Kodaira, Kojima and Yamashita say so in print. The sign $(-1)^{\lambda_1}$ is real, but it speaks about the boundary condition.

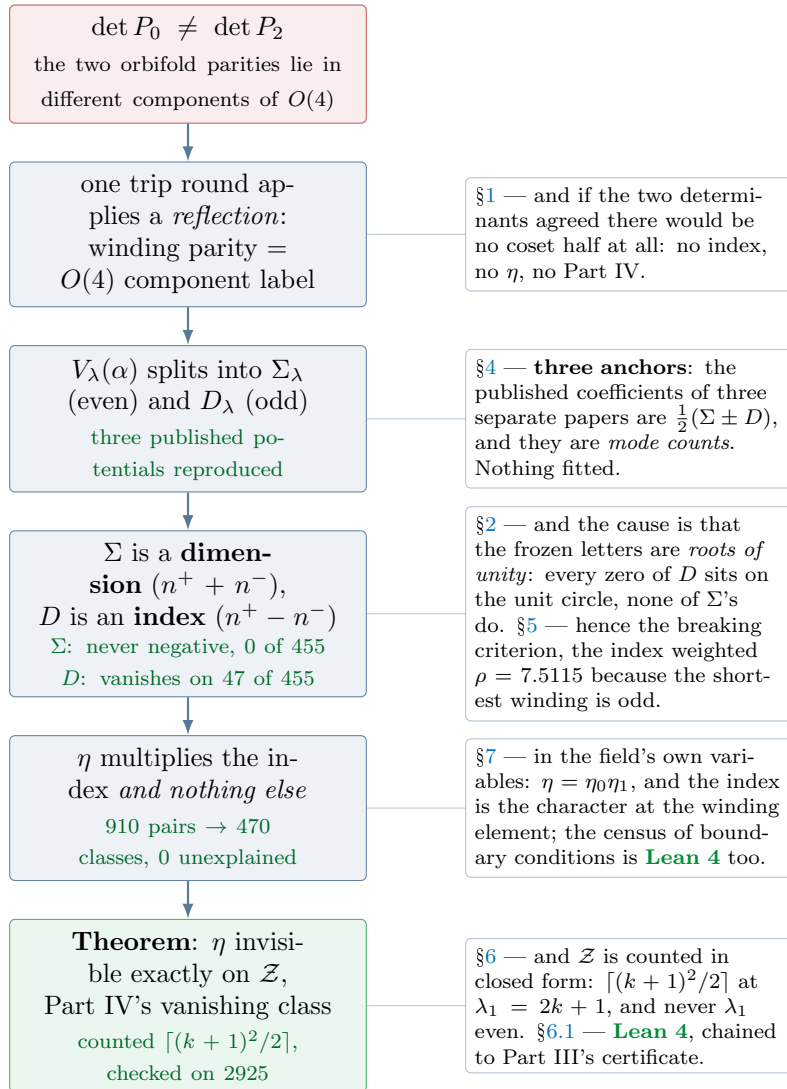


Figure 1: The chain, and it is one chain: every statement in this paper descends from the single determinant at the top. Left, the thread; right, what each link buys and where it is done.

1 The matrix with determinant -1

Part III derived the Kaluza–Klein mode labels of $SU(4)$ gauge-Higgs unification on $T^2/\mathbb{Z}_2$ from the orbifold boundary conditions, and in doing so wrote down its own translation matrix

$$U_6 = P_2 P_0^{-1} = \text{diag}(1, 1, -1, 1), \quad \det U_6 = -1. \quad (1)$$

The determinant is printed there and was never used. It is the hinge of this paper, and it is worth saying at once what acts on what, because every result below is a consequence of one sign.

An orbifold with two fixed points carries two parities, P_0 and P_2 , and the geometry — not a choice — makes the translation their product. Part III’s embedding puts them in *different components* of $O(4)$:

$$P_0 = \text{diag}(1, 1, 1, -1), \quad \det P_0 = -1; \quad P_2 = \text{diag}(1, 1, -1, -1), \quad \det P_2 = +1; \quad (2)$$

so the U_6 of (1) has $\det U_6 = \det P_2 \cdot \det P_0 = -1$: *one trip around the compact direction applies a reflection*. That is the acting object, and what it acts on is the parity grading of the matter multiplet.

Two things have to be said about that determinant before anything is built on it, because both are what a reader should suspect first.

It is not a convention. The parities act on the gauge field by conjugation, and conjugation does not see an overall phase: P and zP act identically. But the orbifold condition is $P^2 = \mathbb{1}$ — stated in exactly that form in [4], their eq. (2.4), $T[P_0]^2 = T[P_1]^2 = I$, and in [5] — and it forces $z^2 = 1$, hence $z = \pm 1$, hence $\det(zP) = z^4 \det P = \det P$. The phase that would flip a determinant in four dimensions is $z = e^{i\pi/4}$, and it gives $(zP)^2 = i\mathbb{1}$, which is not a parity at all. So $\det P$ is pinned by the orbifold condition and is not ours to choose. *On matter* the phase does act, and that is a different statement: it is precisely the boundary sign η of §6. (A setup in which P^2 is only required to lie in the centre would allow $z^8 = 1$ and hence $z^4 = \pm 1$, and the determinant would no longer be fixed; that is not the condition used here or in the sources.)

And it is a statement about the matrix, not about the gauge algebra. Conjugation by any unitary is an *inner* automorphism of $\mathfrak{su}(4)$, since $U = e^{i\theta}V$ with $V \in SU(4)$ conjugates identically; nothing here claims otherwise. What lies in the non-identity component of $O(4)$ is the matrix P_0 itself — real, orthogonal, $\det = -1$ — and that is what the character needs, because s_λ is being evaluated at its eigenvalue alphabet. The determinant enters through the alphabet, not through the automorphism.

The control on (2) is one line and it could have gone the other way. Had the two parities shared a determinant — either both $+1$ or both -1 — the translation would lie in the identity component, every winding with it, and there would be no coset half at all: no index, no boundary-condition sign to be invisible, no vanishing class, and no Part IV. The entire series exists because $\det P_0 \neq \det P_2$.

Everything below is that fact propagating, and Figure 1 is the whole descent on one page: the winding parity becomes the component label (§1); the potential splits into a dimension and an index (§2); the shortest winding is the reflection itself, which is why the index outweighs the dimension (§5); the reflection is not in $SU(4)$, so acting with it on a multiplet requires choosing an extension, and that choice is η (§6); and η is invisible exactly where the reflection acts with balanced multiplicities (Corollary 1).

The (k_1, k_2) winding amplitude of the one-loop potential is the character of the matter multiplet at the holonomy $\exp(i\pi\theta T) U_6^{k_2}$, with $T = \text{diag}(0, 1, 0, -1)$ and $\theta = k_1\alpha_1 + k_2\alpha_2$. Writing $t = e^{i\pi\theta}$, the eigenvalue alphabet of that holonomy is

$$\begin{aligned} k_2 \text{ even: } & (1, 1, t, t^{-1}), \quad \det = +1, \quad \text{identity component,} \\ k_2 \text{ odd: } & (1, -1, t, t^{-1}), \quad \det = -1, \quad \text{reflection coset.} \end{aligned} \quad (3)$$

So the *parity of the winding number is the $O(4)$ component label*. Part IV opened by saying that its alphabet $(1, -1, t, t^{-1})$ “arrived from physics...not from a search of the combinatorial literature”, and could not say why that alphabet and not another. Equation (3) is why: it is the generic semisimple element of the non-identity component, and the physics reaches it because U_6 is not in the identity component. [Verified: 18 of 18 exact matches of the split against Part III’s published histograms, `potential_split.sage`.]

Two scope statements belong here rather than at the end. The construction is $T^2/\mathbb{Z}_2$ with $U_5 = \mathbb{1}$; on $T^2/\mathbb{Z}_4$ the winding sum is a two-element trace $\text{Tr}[W_1^{w_1}W_2^{w_2}]$ and is not a single-holonomy character, so nothing below applies there without redoing it. And the grading of the potential by the centre charge, which one might think is the content of (3), is classical twice over — Davies and McLachlan [13], already cited in Part III, and Roberge–Weiss periodicity [14]; the modern statements of the same grading are in Ünsal–Yaffe and Argyres–Ünsal [15] and Ogilvie [16]. What is offered here is the identification of the two components, not the grading.

The setting itself is old and crowded, which is why every section below states what it inherits. Unifying the Higgs with the extra-dimensional components of a gauge field goes back to Manton [7]; that the resulting Wilson-line phases are dynamical, and break the symmetry by radiative corrections rather than by hand, is the Hosotani mechanism [6]; the orbifold construction that makes four-dimensional chiral fermions possible and splits the Higgs multiplet is due to Kawamura [9] and Hall–Nomura [10], with the supersymmetry-breaking twist of Scherk–Schwarz [11]. The closest five-dimensional computation to the object of this paper — the one-loop potential as a function of the Wilson line phase within one equivalence class — is Haba, Harada, Hosotani and Kawamura [8].

One determinant, and the winding sum splits in two. Into what, exactly — and are the two pieces related, or merely adjacent?

2 The potential is two Schur specialisations

What the Higgs is in this model

Nothing below is about a scalar field added to a Lagrangian. In gauge-Higgs unification the Higgs *is* the Wilson-line phase: the extra-dimensional components A_5, A_6 of the $SU(4)$ gauge field on $T^2/\mathbb{Z}_2$ furnish two four-dimensional doublets, the tree-level potential is $\text{Tr}[A_5, A_6]^2$ and is flat along the Wilson line, and the degeneracy is lifted only at one loop, by the Hosotani mechanism [6]. The vacuum expectation value of the Higgs is therefore a *phase* α , chosen by the minimum of the one-loop potential; the Higgs mass is the curvature of that potential at the minimum; and the couplings κ_V follow from the same function. That is the entire electroweak sector of the model, and it is a sum over windings of the compact directions.

Writing W_k for the (representation-independent) winding weight and $\theta_k = k_1\alpha_1 + k_2\alpha_2$, the one-loop potential of a multiplet λ is exactly

$$V_\lambda(\alpha) = \sum_{k \neq 0} W_k \cdot \begin{cases} \Sigma_\lambda(e^{i\pi\theta_k}), & k_2 \text{ even} & (\text{identity component}), \\ D_\lambda(e^{i\pi\theta_k}), & k_2 \text{ odd} & (\text{reflection coset}). \end{cases} \quad (4)$$

[Verified on all nine representations of the model’s library.] Everything physical in this paper is read off (4): the vacuum is its minimum, the Higgs mass its curvature, and the question of §6 — what a boundary condition can do that anybody could measure — is the question of what can change (4).

The two functions, and why only one of them can vanish

From (3), every winding contributes through one of exactly two functions of the *same* partition λ labelling the multiplet:

$$\Sigma_\lambda(t) = s_\lambda(1, 1, t, t^{-1}), \quad D_\lambda(t) = s_\lambda(1, -1, t, t^{-1}). \quad (5)$$

Part IV computed D_λ and gave a closed form for it: up to a sign it is a product of exactly three $SU(2)$ characters read off the 2-quotient of λ , or identically zero, with no hypothesis on the shape of λ . *Part IV states that identity as verified and not proved* — checked there exhaustively, by two independent implementations, over the ranges it declares, with no mismatch — and every statement in this paper that rests on it inherits that standing exactly; §10 says which ones. The identity sits between two factorisation programmes and belongs to neither: the reciprocal-pair line of Ciucu–Krattenthaler [17] and Ayyer–Behrend [18], and the twining-character line of Jantzen [19] and Kumar–Lusztig–Prasad [20], whose folding identity is the mechanism behind it. In the exponential expansion this makes the coset half a *box*:

$$V_{\text{odd}}(\alpha) = \zeta \sum_{c \in \text{Box}(a_1, a_2, a_3)} G(m(c)\alpha), \quad m(c) = \sum_i a_i - 2 \sum_i c_i, \quad (6)$$

where $\text{Box}(a_1, a_2, a_3) = \{c \in \mathbb{Z}^3 : 0 \leq c_i \leq a_i\}$ and (a_1, a_2, a_3) are the sides read off the 2-quotient, so that $m(c)$ runs over the winding charges of the multiplet with the right multiplicities. The function G in (6) is one and the same for every representation: the multiplet enters only as a multiset of integers read off its Young diagram. That multiset is the list which AHMN [3] (their eq. (3.25)) and Kawamura *et al.* [5] (their eqs. (5.22)–(5.28)) assemble by hand from the Kaluza–Klein spectrum. The curvature of that half — the contribution to the Higgs mass — is its second moment, $\partial_{\alpha_2}^2 V_{\text{odd}} = \zeta \sum_a m(a)^2 G''(m\alpha)$ [verified 9/9 at a generic point, `box.potential.py`], and that second moment is the sum of the three $SU(2)$ Casimirs of the box [verified 2197/2197 over all boxes with sides ≤ 12 , `verify_formulas.py`].

Notation, and it is a collision we have to disarm. Part IV writes the sides of the box as (p, q, r) . The five-dimensional literature we bridge to in §7 writes the block sizes of the orbifold parity matrices as $[p; q, r, s]$, and has done so since 2004. The two triples are unrelated and both are load-bearing here. Throughout this paper the box of Part IV is written (a_1, a_2, a_3) and $[p; q, r, s]$ keeps its established meaning.

Why one half factors and the other cannot

That D_λ always factors and Σ_λ never does is not a difference of degree, and the cause is not that one alphabet has an extra letter. It is that $\{1, -1\}$ is the complete set of square roots of unity and $\{1, 1\}$ is not. Attaching one free reciprocal pair $\{z, z^{-1}\}$ to a frozen block F and asking, by Kronecker’s criterion, whether every zero of s_λ lies on the unit circle — equivalently whether the character is a monomial times a product of cyclotomic polynomials:

frozen block F	non-zero	all zeros on $ z = 1$	share	worst failure
$\mu_2 = \{1, -1\}$	212	212	100%	—
$\{1, 1\}$	241	16	6.6%	1.62
μ_3	124	124	100%	—
$\{1, 1, 1\}$	150	3	2.0%	2.73
μ_4	86	86	100%	—
$\{1, 1, 1, 1\}$	125	2	1.6%	3.79

Only the second column of that table is ours, and the distinction matters. That a root-of-unity block forces a factorisation is classical and has a long line: Littlewood and Richardson showed the Verschiebung action on s_λ vanishes when the t -core is non-empty and otherwise factors over

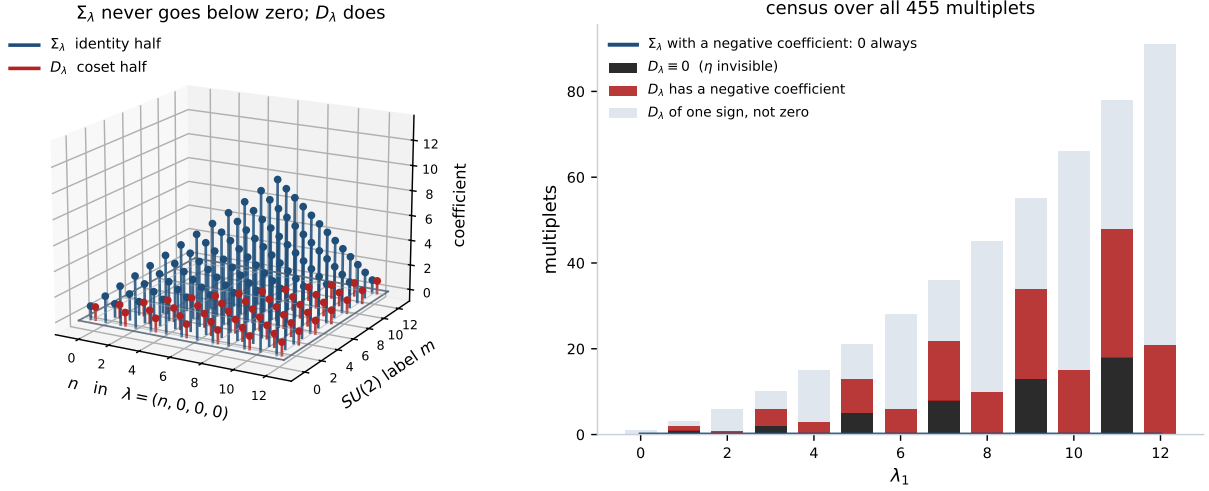


Figure 2: The asymmetry of (8), measured. **Left:** the two halves expanded in $SU(2)$ characters over the symmetric family $\lambda = (n, 0, 0, 0)$, which contains AHMN’s $\mathbf{35} = (4, 0, 0, 0)$. Each stem is one coefficient, and the outlined frame is the zero level. Every blue stem stands above it, because the identity half’s coefficients are dimensions $n_m^+ + n_m^-$; the red ones hang below it, because the coset half’s are indices $n_m^+ - n_m^-$. **Right:** the census over all 455 multiplets in range, by λ_1 and disjointly: those with $D_\lambda \equiv 0$ (the blind class, 47), those whose D_λ has a coefficient of each sign (134), and the rest. The flat line is the count of multiplets whose Σ_λ has a negative coefficient: it is 0 at every λ_1 , and by (8) it must be. Generated by `make_figure_positivity.py`.

the t -quotient; Lecouvey [21] and Ayyer–Kumari [22] did the symplectic and orthogonal cases, characterising non-vanishing by z -asymmetric partitions through a study of beta sets; Albion [23] embeds all of them in one family. *Our* alphabet — all t -th roots of unity together with one free reciprocal pair — is in none of them: that line’s alphabets carry a free variable in every letter, and ours carries exactly one. So the 100% rows are measured here rather than read off a published theorem, and at $t = 2$ they are the zero locus of Part IV’s closed form, with its standing.

What is new here is the **control**: nobody had a reason to run the repeated-letter blocks, because there is no theorem there to prove. We had a reason — $\{1, 1\}$ is the identity half of a Higgs potential — and the answer is that they factor essentially never, and fail grossly. A zero off the circle by 1.62 in modulus is not a numerical artefact, whereas the root-of-unity blocks fail only at a tolerance of 10^{-7} with deviations of order 5×10^{-6} and pass at 10^{-4} [`why_one_factors.py`].

And the division of labour is not the one this series assumed. Four papers went to D_λ because it has zeros to classify and Σ_λ has none, and the sentence above says why. But having no zeros is not the same as carrying no information, and it turns out to be the other way round: Σ_λ alone determines the multiplet up to conjugation — its fibres over 969 multiplets are exactly the orbits $\{\lambda, \lambda^*\}$, with the singletons exactly the self-conjugate ones (§10). So the even-winding half of the potential carries the *identity* of the matter, and the odd-winding half carries the *boundary condition*. The half nobody mined is the one that says what the matter is.

Figure 3 is the same statement counted zero by zero instead of polynomial by polynomial.

Two consequences worth separating. *Mathematically*, “factors into three $SU(2)$ characters” is the right statement only at $t = 2$: at $t = 3$ the pieces are cyclotomic and are *not* $SU(2)$ characters — $\lambda = (2, 1, 0, 0, 0)$ gives $z - 1 + z^{-1}$, the sixth cyclotomic polynomial. The uniform statement across t is the location of the zeros. *Physically*, and this is the reading: *the coset amplitude of a multiplet vanishes only at rational Wilson line phases*, because its zeros are roots of unity in $z = e^{i\pi\theta}$; the identity amplitude has its zeros off the physical line entirely. One half is built to vanish at special points of the Wilson line and the other is not.

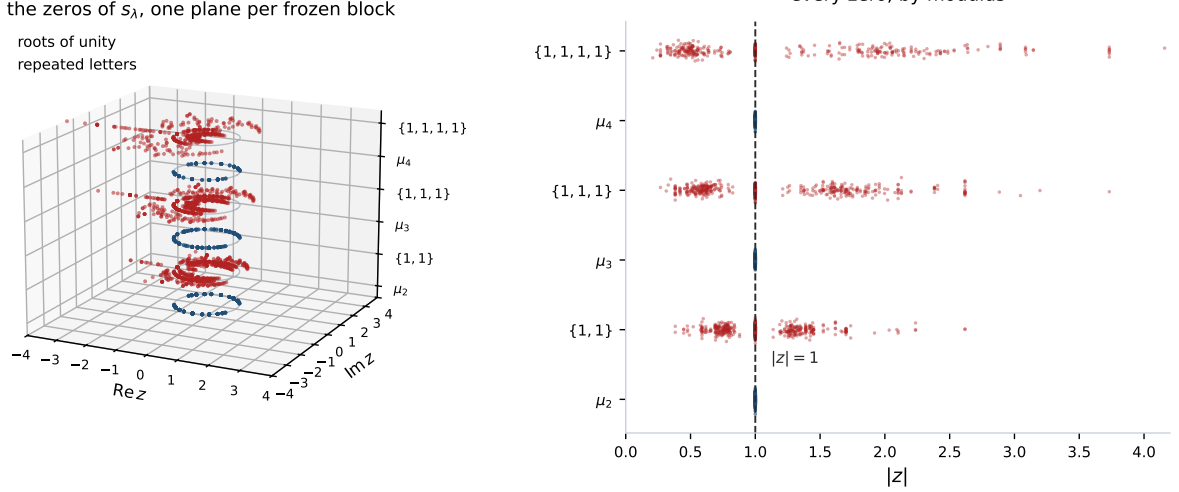


Figure 3: The same dichotomy at the level of individual zeros rather than of percentages. **Left:** the complex z -plane stacked once per frozen block, with the unit circle drawn on each layer. Every zero of every s_λ built on a root-of-unity block lies *on* its circle; every repeated-letter block scatters them off it. **Right:** the same points by modulus. The root-of-unity blocks are a single spike at exactly 1; the others spread to $|z| = 2.618, 3.732$ and 4.791 . Counting zeros rather than polynomials: 618, 408 and 218 zeros for μ_2, μ_3, μ_4 , all at $|z| = 1$ to 10^{-4} ; and 51.6%, 37.5%, 22.5% for $\{1, 1\}, \{1, 1, 1\}, \{1, 1, 1, 1\}$. *The coset half of the potential is built to vanish at rational Wilson line phases; the identity half is not.* Generated by `make_figure_zeros.py`.

The two halves are one operator, traced twice

Before the reading, the identity that makes it exact rather than evocative. Let $\sigma = \text{diag}(1, -1, 1, 1)$ — Part III’s orbifold parity — and let $g_t = \text{diag}(1, 1, t, t^{-1})$ be the Wilson-line element. Then $\sigma g_t = \text{diag}(1, -1, t, t^{-1})$: *the coset alphabet is σ times the identity alphabet*, letter by letter. So the two halves of (4) are

$$\Sigma_\lambda(t) = \text{Tr}_{V_\lambda}(g_t), \quad D_\lambda(t) = \text{Tr}_{V_\lambda}(\sigma g_t), \quad (7)$$

the *trace* and the σ -*twisted trace* of the same group element on the same space [verified: the alphabet identity, and $\frac{1}{2}(\Sigma \pm D)$ non-negative integers at every charge on all multiplets in range, 287 checks, 0 failures, `verify_formulas.py`]. Everything in this paper is a consequence of reading one operator twice. A trace is a dimension and cannot vanish; a twisted trace is an index and can. The boundary sign multiplies the twisted one and nothing else. The blind class is where the grading balances. And the standard form in which this field writes its one-loop potential, $\{A + B(-1)^{k_2}\}$, is that same pair: A and B are the σ -even and σ -odd mode counts, and $A \pm B$ are the trace and the twisted trace. Three separate papers [3, 24, 8] write the twist that exchanges them in three vocabularies — an antiperiodic boundary condition, $Qa \mapsto Qa + \frac{1}{2}$, and $a \mapsto a - \beta$ — and none names it as one operation.

The limit of that language, stated so it is not read for more than it is. “Twisted trace” here is exact bookkeeping and not an index theorem: what classifies the vanishing locus is combinatorics of the 2-quotient and the beta-set (Part IV, and the line from Littlewood–Richardson through [21, 22, 23]), not an analytic index. We claim the identity (7) and the consequences drawn from it, and nothing index-theoretic.

The root of the asymmetry: a dimension and an index

The two halves behave differently, and the reason is not a metaphor about cancellation. Let $\sigma = \text{diag}(1, -1, 1, 1)$ — Part III’s orbifold parity — and let $SU(2)_{34}$ be the subgroup acting on the last two coordinates, whose torus is exactly $\text{diag}(1, 1, t, t^{-1})$. *They commute*, because they act on complementary coordinates. So V_λ decomposes under $SU(2)_{34} \times \langle \sigma \rangle$, and writing $n_m^\pm$ for the multiplicity of the spin- $\frac{m}{2}$ representation inside the $\sigma = \pm 1$ eigenspace, the expansions of the two halves in $SU(2)$ characters χ_m are

$$\Sigma_\lambda = \sum_m (n_m^+ + n_m^-) \chi_m, \quad D_\lambda = \sum_m (n_m^+ - n_m^-) \chi_m. \quad (8)$$

The first coefficient is a *dimension* and the second is an *index*. That is the whole asymmetry, and it is not a choice of language: it is checkable, because it asserts that $\frac{1}{2}(\Sigma_\lambda \pm D_\lambda)$ are non-negative integers. [Verified on all 455 multiplets in range: 0 non-integers, 0 negative. And measured: Σ_λ has a negative $SU(2)$ coefficient in 0 of 455, while D_λ has one in 134 — Figure 2.]

Two things follow, and the second is the one this paper needed.

Corollary 1. $\lambda \in \mathcal{Z}$ if and only if $n_m^+ = n_m^-$ for every m : the vanishing class is exactly the multiplets in which every $SU(2)$ level is parity-balanced.

[Verified: 47 level-balanced, 47 in $\mathcal{Z}$, 0 disagreements.] This refines Part IV, which shows that the *total* parity index $n_+ - n_-$ vanishes on $\mathcal{Z}$. Together with Corollary 2 the chain closes: a single number vanishing — one conserved trace at one fixed point — forces balance at every level of the decomposition.

And it gives η its meaning in one sentence. The boundary sign multiplies the coset half, that is the index; a multiplet whose levels are balanced has index zero at every level, so there is nothing for the sign to multiply. *Invisibility of the boundary condition is exact parity balance of the matter multiplet.*

The same statement in the older language: on weights, the identity component carries $m_\mu + m_{\mu^*}$, which is non-negative so that nothing can cancel, and the coset carries $m_\mu - m_{\mu^*}$, which is signed so that cancellation is possible. Hence:

Even-winding contributions to the Higgs potential can never vanish. Odd-winding ones can, and do, for a set of representations Part IV classifies in closed form.

Part IV’s vanishing criterion therefore has a physical reading: it says *which multiplets contribute nothing to the odd-winding part of the Higgs potential*. Everything in §6 is a consequence of that one sentence, because the boundary sign lives on exactly that part.

The identity half does not collapse to a single box, and the contrast is sharper than a count suggests. Over the 359 partitions with at most four rows and $|\lambda| \leq 16$, exactly 5 have Σ_λ equal to a product of at most three $SU(2)$ characters — and those five are (n, n, n, n) for $n = 0, \dots, 4$, five partition-representatives of the *same* $SU(4)$ irreducible, the singlet, each with $\Sigma_\lambda \equiv 1$. *Non-trivial factorisations: none.* Where D_λ always factors, Σ_λ never does [`verify_formulas.py`, ported from the original Sage check]. In the same block basis, however, it is an explicit *positive superposition* of the same boxes,

$$s_\lambda(1, 1, t, t^{-1}) = \sum_{0 \leq q \leq p \leq \lambda_1} (p - q + 1) B(p, q), \quad B(p, q) = \chi_{m_1} \chi_{m_2} \chi_{m_3}, \quad (9)$$

with $m_1 = \lambda_1 - \max(p, \lambda_2)$, $m_2 = \min(p, \lambda_2) - \max(q, \lambda_3)$, $m_3 = \min(q, \lambda_3) - \lambda_4$, and a term dropped whenever any $m_i < 0$ [verified 286/286 in exact integer arithmetic, `verify_formulas.py`], under the weight $s_\nu(1, 1) = p - q + 1$ where the coset half carries the alternating $s_\nu(1, -1)$.

The summation range in (9) is not a choice, and it is worth saying so because an unstated range is a hidden hypothesis. Since $\max(a, b) + \min(a, b) = a + b$, the three block indices satisfy

$$m_1 + m_2 + m_3 = (\lambda_1 - \lambda_4) - |p - \lambda_2| - |q - \lambda_3| \quad (10)$$

[0 failures in 674 245 pairs]. By (10) the summand is supported exactly where $\lambda_1 \geq \max(p, \lambda_2) \geq \min(p, \lambda_2) \geq \max(q, \lambda_3) \geq \min(q, \lambda_3) \geq \lambda_4$ — an interlacing chain whose first, third and fifth gaps are m_1, m_2, m_3 . That support is finite and the condition $m_i \geq 0$ enforces it by itself: summing over $0 \leq p, q \leq \lambda_1$, over $[-15, \lambda_1+15]^2$ and over $[-40, \lambda_1+40]^2$ gives the same answer, 0 failures of 60 multiplets each; and of the 4104 pairs with all three $m_i \geq 0$ that occur across those three boxes, exactly 0 have $q > p$, so even $q \leq p$ is implied rather than imposed. There is no range constant to get right.

The *positivity* of that expansion is not a finding and we do not present it as one: $\text{diag}(1, 1, t, t^{-1})$ is the torus of an $SU(2)$ subgroup, so Σ_λ is a restriction of V_λ to that subgroup and its multiplicities are non-negative for the reason all multiplicities are. What is ours is the explicit weight $p - q + 1$ and the block indices it runs over. So the whole one-loop potential, both halves, is a stack of boxes built from the one universal G ; the coset half collapses to a single box because its weights alternate, and the identity half does not because its weights are positive.

This is worth stating as a physical claim and not only as a symmetric-function identity, because it is the operational content of the whole construction:

For any multiplet, the entire one-loop Higgs potential (4) — both halves — is a positive-integer-weighted sum of one universal function evaluated at integer charges read off the Young diagram, the coset half carrying one global sign ζ in front of it. No Kaluza–Klein spectrum has to be enumerated.

The claim has to be narrowed, and here is the narrowing. A general one-loop potential for this kind of model is not new: Haba and Yamashita [24] give one for 5D $SU(N)$ on $S^1/\mathbb{Z}_2$ in 2004, with explicit formulas for the adjoint, the fundamental and the (anti)symmetric tensors and a method they state extends to higher representations. What is not in their paper, nor in [3] whose eq. (3.25) and [5] whose eqs. (5.22)–(5.28) are lists assembled *per representation* from the mode expansion, is a formula *closed in λ* : one universal function whose integer charges are read off the Young diagram of *any* irreducible, with no decomposition performed representation by representation. That, and only that, is the claim here. The consequence is that the vacuum $\alpha_{\min}$, the curvature at it, and κ_V become computable for *arbitrary* matter content rather than for the handful whose spectra someone has written out. What one then *does* with that — run it forwards to predict a spectrum — is Part VI’s business, not this paper’s.

Two honest limits on that claim, stated here and not later. The identity stack may still have a shorter description; we do not have one. And a closed form for the *potential* is not yet a closed form for the *Higgs mass*: the mass is the curvature at the true minimum, and only the curvature at the origin is available in closed form so far. Mining the identity half for the leading behaviour of m_h is open, and it is the piece of this series we have looked at least.

Three pairs have now appeared and all three transform the same way: the two $O(4)$ components, the two eigenspaces of the associate involution, and the two orbifold mode parities. Three coincidences, or one object seen three times?

3 One $\mathbb{Z}_2$, seen three times

Before the anchor, it is worth saying what the objects of §2 have in common, because they are not three coincidences. Every formula in this paper is the same $\mathbb{Z}_2$ Fourier transform — $\{f_+, f_-\} \mapsto \{f_+ + f_-, f_+ - f_-\}$ — read at three different levels, and the content of the paper is that the three agree.

level	the $\mathbb{Z}_2$	what the transform gives
the group	$\pi_0(O(4))$: identity component vs. reflection coset	Σ_λ and D_λ , selected by the parity of the winding
the multiplet	$\sigma = \text{diag}(1, -1, 1, 1)$ acting on V_λ	$\Sigma_\chi = n^+ + n^-$ (a dimension) and $D_\chi = n^+ - n^-$ (an index), (8)
the spectrum	the orbifold parity of a Kaluza–Klein mode	$A = \frac{1}{2}(\Sigma + D)$ and $B = \frac{1}{2}(\Sigma - D)$, the published coefficients of §4

The third row is the one that turns the anchor from an agreement of two lists into a statement with a mechanism. AHMN obtain their A and B by counting Kaluza–Klein modes of each orbifold parity at each charge. Our claim is that Σ_λ and D_λ are the *generating functions* of exactly that count, so A and B are not fitting constants: they are mode numbers. That is a falsifiable arithmetic statement — $\frac{1}{2}(\Sigma \pm D)$ must be *non-negative integers* at every charge, for every multiplet — and it holds:

check over the 455 multiplets in range	failures
$A = \frac{1}{2}(\Sigma + D)$ a non-negative integer at every charge	0
$B = \frac{1}{2}(\Sigma - D)$ a non-negative integer at every charge	0
$\frac{1}{2}(\Sigma_\chi \pm D_\chi)$ non-negative integers (the multiplet level)	0
D_λ with $SU(2)$ coefficients of both signs	0

The last row is a consequence of Part IV and it is the sharpest of the four. Part IV’s closed form gives $D_\lambda = \zeta \chi_{a_1} \chi_{a_2} \chi_{a_3}$ or 0; a product of $SU(2)$ characters has non-negative Clebsch–Gordan multiplicities; therefore

$$n_m^+ - n_m^- = \zeta \cdot c_m, \quad c_m \geq 0 \text{ for every } m. \quad (11)$$

The parity imbalance of a multiplet points the same way at every $SU(2)$ level, and ζ — which Part IV identified as the sign of the total parity index — is that single direction. Corollary 1 is then immediate rather than surprising: by (11) a sum of terms of one sign vanishes only if every term does, so a vanishing total index forces balance level by level. Corollary 2 is the same argument in the other basis.

————— *If the third row of that table is right, then somebody else’s published coefficients — got by counting modes, not by any of this — have to be our $\frac{1}{2}(\Sigma \pm D)$. That is a claim about numbers we did not produce, and it is the next thing to check.*

4 Three anchors: published potentials, reproduced

A structural claim of this kind is worth nothing until it reproduces numbers someone else printed. One such agreement is a coincidence one cannot rule out, so there are three, chosen to share as little as possible. Akamatsu, Hirose, Maru and Nago [3] compute the one-loop potential of this model and write it with coefficients of the form $\{A + B(-1)^{k_2}\}$. Under (3) that form is forced, and the dictionary is fixed with no freedom: k_2 even selects $A + B$ and k_2 odd selects $A - B$, so

$$A = \frac{1}{2}(\Sigma_\lambda + D_\lambda), \quad B = \frac{1}{2}(\Sigma_\lambda - D_\lambda). \quad (12)$$

Evaluated per charge (the exponent halved) and compared against their printed values, with nothing fitted:

charge	ours		ours		AHMN eq. (3.25)	
	Σ	D	A	B	A	B
2.0	1	1	1	0	1	0
1.5	2	0	1	1	1	1
1.0	4	2	3	1	3	1
0.5	6	0	3	3	3	3

Eight numbers for the $\mathbf{35} = (4, 0, 0, 0)$, exact. Their eq. (3.11), the gauge adjoint $\mathbf{15} = (2, 1, 1, 0)$, matches after the $(-1)^{k_2}$ twist *they* state in their text (antiperiodic boundary conditions), which under (12) is precisely $A \leftrightarrow B$: raw $\{2 : (0, 1), 1 : (2, 2)\}$ becomes $\{2 : (1, 0), 1 : (2, 2)\}$. Their eq. (4.1) is their own $3 \times (3.25) - (3.11)$, re-derived here as an internal control. Free of charge, the two dimension checks $\Sigma(1) = 35$ and $\Sigma(1) = 15$ come out right. Twelve published numbers, nothing fitted [ahmn_anchor.py, output in outputs/ahmn_anchor.txt].

Read with §3, the table says more than that the numbers agree: the column A is the count of parity-even modes at that charge and B the count of parity-odd ones, which is how they were obtained in the first place. The dictionary (12) is the inverse $\mathbb{Z}_2$ transform, and the reason it needs nothing fitted is that there is nothing to fit.

A second anchor, chosen to be as different as we could find

One anchor is a coincidence one cannot rule out. The second is Haba and Yamashita [24]: a different group, a different number of dimensions, a different orbifold and a different derivation — they decompose the representation under the $SU(2)$ picked out by the Wilson-line VEV and read off the generator eigenvalues, where we read a Young diagram. Their worked example is $SU(3)$ on $S^1/\mathbb{Z}_2$ with $P = P' = \text{diag}(+, -, -)$, and they print four things. All four come out of our mode counting with nothing adjusted [hy_anchor.py]:

their equation	what it says	ours
(3.8)	charges $(+1, -1, 0, 0, +\frac{1}{2}, -\frac{1}{2}, +\frac{1}{2}, -\frac{1}{2})$	identical multiset
(3.2)	every basis element is $(+, +)$ or $(-, -)$, never mixed	identical
(3.9)	modes: 2 at $ Q = 1$, 4 at $ Q = \frac{1}{2}$, 2 at $Q = 0$	identical
(3.10)	$V = \frac{C}{2} \sum_n n^{-5} [\cos 2na + 2 \cos na]$	$[\cos 2na + 2 \cos na]$

And their eq. (3.11) is worth a sentence of its own. For $\eta = -1$ they shift $Qa \mapsto Qa + \frac{1}{2}$, which is a factor $(-1)^n$ on the winding sum — that is $A \leftrightarrow B$ in the notation of §3, and it is *the same twist* AHMN state for their gauge sector as an antiperiodic boundary condition. Two independent papers, two vocabularies, one operation.

One sentence of method, because it nearly went in wrong. Compared naively — our characters against their coefficients, term by term — the numbers do *not* agree, 0 of 4. It is (12) that makes them agree, and (12) is the physical claim, not a fitting convenience. A quick comparison would have reported a refutation.

The dictionary holds against the printed numbers of three separate papers. What it does not yet say is which of the two halves can block the breaking of electroweak symmetry — and only one of them can, for a reason that is arithmetic rather than dynamical.

5 The two halves, and which one can block

The two specialisations are the eigenspaces of the $O(N)$ associate involution $\mu \mapsto \mu^*$ on weights: the identity component carries the symmetric combination $m_\mu + m_{\mu^*}$, which is non-negative and cannot cancel, and the coset carries the antisymmetric $m_\mu - m_{\mu^*}$, which can. The physical reading is immediate and is the reason Part IV’s vanishing criterion matters here: *the identity half always contributes with the symmetry-breaking sign to the curvature at the origin, and the coset half is*

the only thing that can cancel it there — a statement about the origin, not about $V(\alpha)$ globally, as the caveat below (13) makes explicit. With the geometric weight $\rho = 7.5115$ for 6D on T^2 , the closed criterion

$$\text{breaks} \iff \sum_i n_i \left[\frac{2 \dim_i C_2(i)}{\dim G} + \rho \varepsilon_i M_2 \delta_i \right] > 0 \quad (13)$$

reproduces the direct computation.

Why the index outweighs the dimension, and by how much

Criterion (13) is a competition between a dimension and an index, and the index enters weighted $\rho \approx 7.5$ times more per unit moment. That factor is not a coupling and it is not fitted; it is geometry, and its origin is one line of arithmetic.

The curvature in α_2 weights each winding k by k_2^2 , so the winding $k_2 = 0$ contributes *exactly nothing* — the identity sector loses its shortest shell for free. Its next shell is $|k_2| = 2$, twice as long, while the coset sector starts at $|k_2| = 1$; and the winding sum falls off like $|k|^{-2p}$, with $p = 3$ for 6D. So the coset sector owns the shell that dominates the sum:

shell $ k_2 $	contribution at $p = 3$	
0	0.00000	even — absent, killed by the k_2^2 weight
1	2.53718	odd — 95% of the whole coset sum
2	0.29466	even — 83% of the whole identity sum
3	0.08727	odd

so $\rho = A_{\text{odd}}/A_{\text{even}} = 2.65906/0.35400 = 7.5115$. The control that could have falsified this reading does the opposite: removing *only* the $|k_2| = 1$ shell sends $\rho \rightarrow 0.3443$, and the index stops outweighing the dimension altogether. And ρ is a pure function of the fall-off, hence of the dimension of the compactification: 1.0060 at $p = 1.5$, 7.5115 at $p = 3$, 35.5608 at $p = 4$, 676.8241 at $p = 6$.

The index weighs more because the shortest winding is odd. That is the whole of it, and it means ρ is fixed by the geometry rather than chosen. It also sets a limit on how much weight (13) can carry: the base **60** of this model has critical $\rho = 6.4545$ against an actual 7.5115, so a 14.1% shift in ρ reverses its verdict. Verdicts near that margin should be read as such.

A caveat that belongs next to (13) and not in a footnote: it tests the instability of the *origin*. A content can have a stable origin and a lower minimum elsewhere — the **45** does — so the criterion is local, not global.

Part IV's vanishing class $\mathcal{Z}$ therefore names exactly the matter that has no say in blocking: such a multiplet contributes through its Dynkin index and nothing else.

————— *So one half of the potential can vanish outright, on a class already classified in closed form. The question this paper exists to answer is what stops being observable when it does.*

6 What the potential can see

The sign, and what it is

Each matter multiplet carries a discrete boundary-condition sign. In our variables it is the determinant twist $\lambda \mapsto \lambda + (1, 1, 1, 1)$, which is the *same* $SU(4)$ irreducible representation [verified on 212 partitions] and multiplies the character by det: trivial on the identity component, a sign on the coset. So $\eta \in \{\pm 1\}$ acts as

$$\eta : (\Sigma_\lambda, D_\lambda) \longmapsto (\Sigma_\lambda, \eta D_\lambda), \quad (14)$$

and the physical observable of a multiplet is the *pair* $(\Sigma_\lambda, \eta D_\lambda)$: two multiplets with the same pair give the same potential at every α , hence the same vacuum, the same masses and the same κ_V .

That sign is not ours and it has a name. Haba, Hosotani and Kawamura [4] attach to each matter field two signs $\eta_0, \eta_1 \in \{\pm 1\}$, one per orbifold parity, $\phi(x, -y) = \eta_0 P_0 \phi(x, y)$ and $\phi(x, R - y) = \eta_1 P_1 \phi(x, R + y)$; the overall flip $(\eta_0, \eta_1) \rightarrow (-\eta_0, -\eta_1)$ is a redundancy they state themselves, and everything finite in the potential depends on the pair through the product alone. Our η is that product,

$$\eta := \eta_0 \eta_1, \quad (15)$$

and §7 makes the identification exact rather than terminological.

The theorem

Theorem 1. *For every pair (λ, η) ,*

$$(\lambda, \eta) \sim (\lambda^*, \eta(-1)^{\lambda_1}) \quad (16)$$

is an observational identity: the two pairs give the same $(\Sigma_\lambda, \eta D_\lambda)$ and therefore the same potential at every α . And η is unobservable for λ if and only if $D_\lambda \equiv 0$ — by Part IV’s identity, exactly on its vanishing class $\mathcal{Z}$: for those multiplets the boundary condition has no consequence that any measurement of this Higgs sector can see.

Proof of the relation. Both alphabets in (3) are stable under inversion, so $s_\lambda(A^{-1}) = s_\lambda(A)$; λ^* is the complement of λ in the $4 \times \lambda_1$ rectangle, so the classical complementation identity gives $s_{\lambda^*}(A) = \det(A)^{\lambda_1} s_\lambda(A)$; and $\det = +1$ on the identity component, -1 on the coset. Hence $\Sigma_{\lambda^*} = \Sigma_\lambda$ always and $D_{\lambda^*} = (-1)^{\lambda_1} D_\lambda$, which is (16) acting on (14). That η is unobservable when $D_\lambda \equiv 0$ is (14) itself. $\square$

The theorem says that (16) is a collision and when it degenerates. That it is the *only* one is a separate statement with a different standing, and it is stated separately:

Observation 1 (completeness — verified, not proved). *Over the range swept, every observational collision is an instance of (16): no two pairs outside the relation share a $(\Sigma_\lambda, \eta D_\lambda)$.*

The sweep is over $SU(4)$ irreducibles with $\lambda_1 \leq 12$ and $\lambda_4 = 0$, in exact integer Laurent arithmetic [fibre_eta.py, data in outputs/fibre_eta.json]:

pairs (λ, η)	910
observational classes	470
fibre sizes	56 of size 1, 401 of size 2, 13 of size 4 — nothing else
fibres not generated by (16)	0
irreducibles in $\mathcal{Z}$	47
irreducibles whose η is unobservable	47, the same ones, by no other mechanism

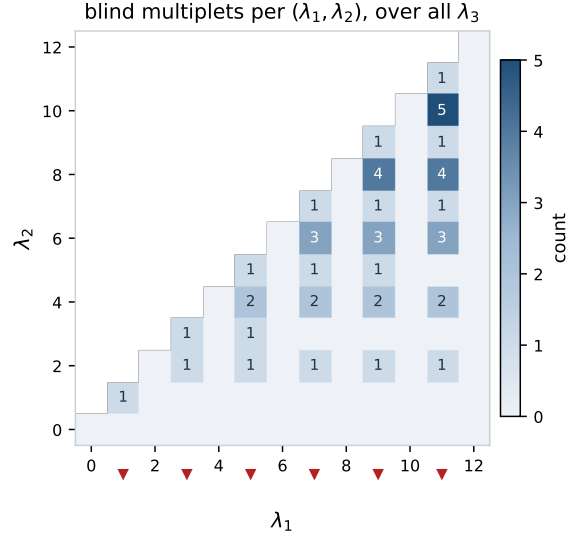
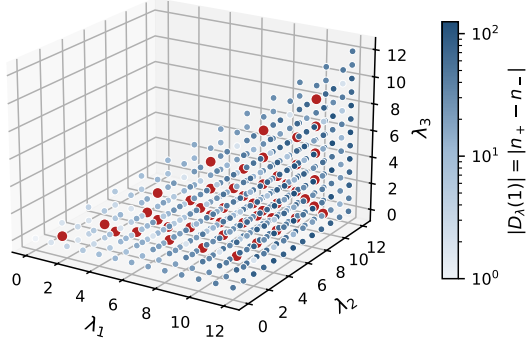
Figure 5 is the three cases side by side. The classes of size four are the 13 conjugate pairs inside $\mathcal{Z}$, where both $\lambda \leftrightarrow \lambda^*$ and $\eta \leftrightarrow -\eta$ act and $\{(\lambda, \pm), (\lambda^*, \pm)\}$ collapses to a single observable.

Why the two mechanisms are one

Relation (16) degenerates — sending (λ, η) to $(\lambda, -\eta)$ — exactly when λ is self-conjugate with λ_1 odd. With $\lambda_4 = 0$, self-conjugate means $\lambda_2 + \lambda_3 = \lambda_1$, that is $\lambda_i + \lambda_{5-i} = \lambda_1$ for every i : self-complementary of width λ_1 . With λ_1 odd that is, word for word, branch (b) of Part IV’s vanishing criterion. So the degeneracy of the relation is not extra information; it is the same statement seen twice [verified: 21 of 21 such irreducibles lie in $\mathcal{Z}$, out of 49 self-conjugate ones in range].

the 455 multiplets, coloured by $|D_\lambda(1)|$

• $D_\lambda \equiv 0$: η invisible (47 of 455)



λ_1 odd: the only columns holding a blind multiplet

Figure 4: The blind class, drawn from the archived sweep. **Left:** every $SU(4)$ multiplet with $\lambda_1 \leq 12$, $\lambda_4 = 0$, placed at $(\lambda_1, \lambda_2, \lambda_3)$ and coloured by the magnitude $|D_\lambda(1)| = |n_+ - n_-|$, the parity index of Part IV, on a logarithmic ramp of one hue. The 47 multiplets of $\mathcal{Z}$ carry no magnitude at all and are drawn as their own class: these are the ones whose boundary-condition sign η no measurement of this Higgs sector can see. **Right:** the same set projected on (λ_1, λ_2) , each cell counting the λ_3 that give a blind multiplet. The class is a structured surface, not scattered dust — and the structure is visible without any theory: *no multiplet with λ_1 even is ever blind* (§6). Generated by `make_figures.py` from `outputs/fibre_eta.json`; nothing is recomputed for the figure.

How big is the blind class

Because Part IV classifies $\mathcal{Z}$ in closed form, the blind set can be counted rather than estimated, and the count has more structure than we expected. For fixed λ_1 the sweep is complete — λ_2 and λ_3 range over everything allowed — so these are exact counts and not samples:

λ_1	1	3	5	7	9	11	any even λ_1
blind multiplets	1	2	5	8	13	18	0
$\lceil (k+1)^2/2 \rceil, \lambda_1 = 2k+1$	1	2	5	8	13	18	—

Both statements are consequences of Part IV’s criterion, and reading them off it is a short argument rather than a debt:

Proposition 1. *Let $\lambda = (\lambda_1, \lambda_2, \lambda_3, 0)$. Then $\lambda \in \mathcal{Z}$ if and only if*

- (a) λ_1 odd, λ_2 even, λ_3 odd; or
- (b) λ_1 odd, λ_2 odd, λ_3 even and $\lambda_2 + \lambda_3 = \lambda_1$.

The two cases are disjoint. In particular λ_1 is odd for every $\lambda \in \mathcal{Z}$, and for $\lambda_1 = 2k+1$

$$\#\{\lambda \in \mathcal{Z}\} = \underbrace{\frac{k(k+1)}{2}}_{(a)} + \underbrace{\lfloor k/2 \rfloor + 1}_{(b)} = \left\lceil \frac{(k+1)^2}{2} \right\rceil.$$

Proof. The beta-set is $\beta = (\lambda_1 + 3, \lambda_2 + 2, \lambda_3 + 1, 0)$, and Part IV gives $D_\lambda \equiv 0$ in exactly two situations: one parity class of β is empty, or the split is $2 + 2$ and $d_3 = |a_1 + a_2 - b_1 - b_2| = 0$, where $\chi_{-1} = 0$.

Since $\beta_4 = 0$ is even, the even class is never empty, so the first situation is that the odd class is empty: $\lambda_1 + 3, \lambda_2 + 2, \lambda_3 + 1$ all even, which is (a).

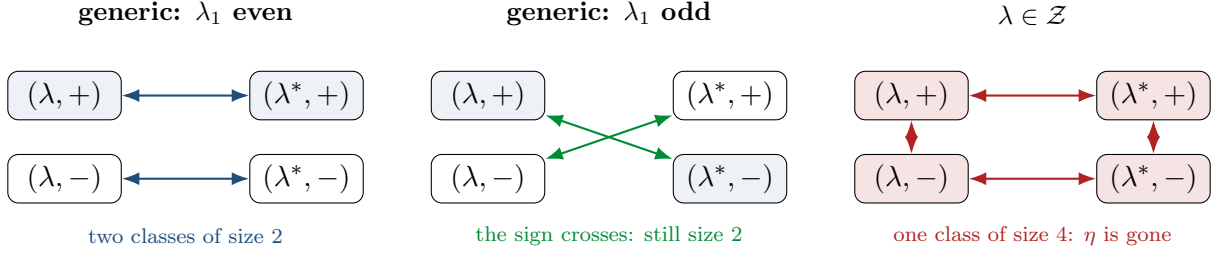


Figure 5: The fibre of $(\lambda, \eta) \mapsto (\Sigma_\lambda, \eta D_\lambda)$, which is the observational equivalence class. Shaded boxes lie in one class. The sign η survives as an observable in the first two panels and disappears in the third, and the third is exactly $\mathcal{Z}$: $D_\lambda \equiv 0$ makes η act trivially there, on all 47 multiplets of the class. A second route reaches the same place — when λ is self-complementary of odd width, (16) degenerates and identifies $(\lambda, +)$ with $(\lambda, -)$ by itself — and it is not independent: those are 21 multiplets and all 21 lie inside the 47. No multiplet is blind to η by any other mechanism. Counts over $\lambda_1 \leq 12$: 56 + 401 + 13 classes of size 1, 2, 4 covering all 910 pairs.

For the second, the even class is $\{0, x\}$ with x the unique even entry among $\beta_1, \beta_2, \beta_3$. If $x = \beta_1$ then λ_1 is odd, the odd class is $\{\lambda_2 + 2, \lambda_3 + 1\}$, and $d_3 = 0$ reads $\lambda_1 + 3 = \lambda_2 + \lambda_3 + 3$, which is (b). If $x = \beta_2$ then $d_3 = 0$ reads $\lambda_2 = \lambda_1 + \lambda_3 + 2 > \lambda_1$, contradicting $\lambda_2 \leq \lambda_1$; if $x = \beta_3$ it reads $\lambda_3 = \lambda_1 + \lambda_2 + 4 > \lambda_3$, impossible. The cases are disjoint because λ_2 is even in (a) and odd in (b).

Counting at $\lambda_1 = 2k + 1$. In (a), $\lambda_2 = 2j$ with $0 \leq j \leq k$, and λ_3 odd with $\lambda_3 \leq 2j$ leaves $\lambda_3 \in \{1, 3, \dots, 2j - 1\}$, that is j values; summing gives $k(k + 1)/2$. In (b), $\lambda_3 = \lambda_1 - \lambda_2$ is determined and automatically even, and $\lambda_2 \geq \lambda_3$ is $2\lambda_2 \geq 2k + 1$, i.e. $\lambda_2 \geq k + 1$; the odd integers in $[k + 1, 2k + 1]$ number $(k + 1) - \lceil k/2 \rceil = \lfloor k/2 \rfloor + 1$. Finally $k(k + 1)/2 + \lfloor k/2 \rfloor + 1 = \lceil (k + 1)^2/2 \rceil$, by writing $k = 2m$ and $k = 2m + 1$ separately. $\square$

Figure 4 draws the class the proposition counts, and its right-hand panel is the parity statement made visible: the populated columns are exactly the odd ones.

[Verified independently of the proof, on all 2925 multiplets with $\lambda_1 \leq 24$: the characterisation disagrees with the exact computation of D_λ in 0 cases, the two branches are disjoint, and each branch total is predicted separately — branch (a) 286 against 286 measured, branch (b) 42 against 42.]

6.1 The proposition is machine-checked, and it descends from Part III's certificate

Proposition 1 is certified in Lean 4 [29] (`GHUBlindCount.lean`, Table 1). The point is not that a counting argument has been re-typed in a proof assistant; it is *where the certificate starts*. Part III already carries a machine-checked brick, `NotchCentreCharge.lean`, whose main theorem is the vanishing criterion in Dynkin labels,

$$N a b c = 0 \iff \text{Odd } b \wedge ((\text{Odd } a \wedge \text{Odd } c) \vee a = c),$$

proved by an honest Laplace expansion over $\mathbb{Z}[T; T^{-1}]$ with no `decide` on the determinant. Part V's proposition is stated in partition coordinates, and the passage between the two is the substitution $a = \lambda_1 - \lambda_2$, $b = \lambda_2 - \lambda_3$, $c = \lambda_3$. That substitution is exactly the step at which a paper of this kind silently goes wrong, so it is the step we certified first: `blind_iff_N_zero` closes the loop, and the count is therefore a theorem *about the determinant*, not about a predicate we invented to be countable.

It also settles, as a by-product, something the series had never checked mechanically. The proof of Proposition 1 above reads the two branches off *Part IV's* criterion — the parity classes of the beta-set and the 2+2 split with $d_3 = 0$ — while `blind_iff_notch` proves those same two branches equivalent to *Part III's* criterion in Dynkin labels. The two papers therefore describe

Lean theorem (GHUblindCount.lean)	content
blind_iff_notch	the dictionary: on $\lambda_1 \geq \lambda_2 \geq \lambda_3$, branches (a)/(b) are the Dynkin criterion
blind_iff_N_zero	$\lambda \in \mathcal{Z} \leftrightarrow N(\lambda_1 - \lambda_2, \lambda_2 - \lambda_3, \lambda_3) = 0$ — the bridge to NotchCentreCharge.lean
blind_l1_odd	$\lambda \in \mathcal{Z} \Rightarrow \lambda_1$ odd; no even- λ_1 multiplet is ever blind
card_shellA, card_shellB	branch (a) = the strict pairs $q < p \leq k$, branch (b) = the interval $m \leq \lfloor k/2 \rfloor$; (a) is stated doubled, $2 \#(a) = (k+1)k$, so no division on $\mathbb{N}$ enters the proof
card_shell	$\#\{\lambda \in \mathcal{Z} : \lambda_1 = 2k+1\} = \lceil (k+1)^2/2 \rceil$ — Proposition 1

Table 1: The machine-checked core of Part V, matter side. All statements are `sorry`-free and depend only on `propext`, `Classical.choice`, `Quot.sound`. The parity of λ_2 is what splits the shell into the two branches, so the case division of the paper proof and the case division of the Lean proof are the same one. The geometry side — the census of §7, Proposition 4 — is a second file, `GHUCosetCensus.lean`, with the same standing and the same three axioms.

one set, and the identification is now machine-checked rather than assumed. It had been assumed since Part IV.

Before a line of Lean was written the dictionary was swept in Python over the 39711 partitions with $\lambda_1 \leq 60$, with 0 disagreements — and, because a control that cannot fail is not a control, the same sweep was run against a deliberately corrupted dictionary ($c := b$), which it rejects on 5200 partitions.

Where the certificate stops, exactly. The chain is machine-checked from the bialternant determinant onward. It is *not* machine-checked back to the character: the identification $D_\lambda(t) = s_\lambda(1, -1, t, t^{-1}) = N/V$ is the classical bialternant definition of a Schur function, used here as a classical input, together with $V \neq 0$ (which *is* certified, as `V.ne_zero`). That boundary is not a matter of effort: Mathlib [30] has no Schur functions at all — it carries `YoungDiagram` and `SemistandardTableau`, and its three files named for Schur are Schur’s lemma, Schur–Zassenhaus and the Schur complement. Formalising $s_{\lambda^*}(A) = \det(A)^{\lambda_1} s_\lambda(A)$, the complementation identity on which Theorem 1 rests, would require building that theory first, and we do not claim it. Theorem 1 is therefore proved but not certified; Proposition 1 is both.

One step past a single multiplet, and it is free

The certificate stops there; the *statement* does not stop at a single multiplet, and the step past it costs one line. A physical content is a multiset $\{(\lambda_i, \eta_i, n_i)\}$ with multiplicities $n_i > 0$ and the observable is the sum, so two contents could still collide by a cancellation that no pair of multiplets performs. Half of that possibility closes in one line. Write ζ_i for the sign of Part IV’s closed form at λ_i , and call $\eta_i \zeta_i$ the *effective sign* of that multiplet.

Proposition 2 (coherent signs). *If every non-blind multiplet of a content carries the same effective sign, then $\sum_i n_i \eta_i D_{\lambda_i} \equiv 0$ if and only if every λ_i is blind on its own.*

Proof. Part IV’s closed form gives $D_{\lambda_i} = \zeta_i P_i$ with P_i of non-negative coefficients, so $\eta_i D_{\lambda_i} = (\eta_i \zeta_i) P_i$. If the effective signs share one value ε the sum is $\varepsilon \sum_i n_i P_i$, a sum of terms of one sign with positive weights, and it vanishes only if every P_i does. $\square$

So **collective blindness requires sign frustration**: a content that is not blind term by term can only go blind if it carries *both* values of $\eta_i \zeta_i$. This does not classify the frustrated case — that

is an integer kernel problem and it is the next paper — it removes the half that is structurally simple, and it carries the standing of Part IV’s identity, on which the one-sign property rests.

What this section claimed in an earlier draft, and why it was wrong

The first version of this section said: *the coset half is the only observable in this model that can see charge conjugation*. That is false, and it is contradicted in print. Kawamura, Kodaira, Kojima and Yamashita [5] write of the conjugate representation, in their §5.1, “which is equivalent with the one of 4. This holds for any representation” — and in their formalism it is one line, since their potential is a sum of cosines and conjugation negates the charges. Charge conjugation is invisible here *unconditionally*.

The sign $(-1)^{\lambda_1}$ in (16) is real; what it speaks about is η . Because U_6 has $\det = -1$ it does not lie in $SU(4)$, so “the character of $\bar{R}$ at U_6 ” is not defined by $SU(4)$ alone: it depends on which extension of the multiplet to the group generated by $SU(4)$ and the determinant-flipping element is chosen, and the two natural extensions differ by exactly $\det^{\lambda_1}$. That choice is the boundary condition. The statement that survives is Theorem 1, and it is sharper than the one that died.

The theorem is phrased in our own machinery, which is exactly the condition under which a claim has not been gated at all. Said in the language the field has used since 2004, does it survive — and does it stay a statement about one six-dimensional model?

7 The same statement in the field’s own variables

Theorem 1 is phrased in our machinery. It need not be, and the translation does more than rename things: it shows the statement is a property of the five-dimensional classification of boundary conditions and not of the particular model of §4.

Haba, Hosotani and Kawamura [4] classify the boundary conditions of $SU(N)$ gauge theory on $S^1/\mathbb{Z}_2$ by the block sizes $[p; q, r; s]$ of the two commuting parity matrices P_0, P_1 , prove that every equivalence class has a diagonal representative, and count the classes: $(N+1)^2$. For the fundamental, the second-rank antisymmetric and the adjoint they tabulate (their eq. (3.20)) the multiplicities $N_{\text{rep}}^{(\varepsilon_0, \varepsilon_1)}$ of basis vectors by their (P_0, P_1) parity. Those tables are three characters in disguise:

Proposition 3. *For any representation of $SU(N)$ and any commuting pair of parities (P_0, P_1) ,*

$$N^{(\varepsilon_0, \varepsilon_1)} = \frac{1}{4} \left[\dim + \varepsilon_0 \chi(P_0) + \varepsilon_1 \chi(P_1) + \varepsilon_0 \varepsilon_1 \chi(P_0 P_1) \right]. \quad (17)$$

Figure 6 draws these three characters over the whole space of boundary conditions, and shows that each layers it in a different direction. Equation (17) is the trace of the projector $\frac{1}{4}(1 + \varepsilon_0 P_0)(1 + \varepsilon_1 P_1)$, and it is verified against their printed table on every $[p; q, r; s]$ up to $N = 8$: 1470 cases, 0 disagreements with eq. (3.20) and 0 with (17) [hhk.bridge.py, output in outputs/hhk_bridge.txt]. It is elementary, and it is recorded here as a bridge and not as a discovery: what it buys is that the three numbers on the right are the *only* way a boundary condition enters, for every representation and not only for the three that are tabulated.

Those three numbers are not new either, and the reader should meet their owners here. Takeuchi and Inagaki [26, 27] observed that a boundary-condition-connecting gauge transformation acts on the rotation matrix at a fixed point by conjugation with Ω *evaluated at that fixed point* — a constant — so that the trace there is invariant by cyclicity, and they call the resulting identities *trace conservation laws*. With them they classify the equivalence classes on $S^1/\mathbb{Z}_2$, $T^2/\mathbb{Z}_3$ and $T^2/\mathbb{Z}_m$ ($m = 2, 3, 4, 6$) without ever writing a gauge transformation, recovering the counts and exhibiting off-diagonal classes. Their argument is representation-blind, so χ_R at each

the 35 boundary conditions $[p; q, r; s]$ of $SU(4)$, seen by the multiplet $\mathbf{15} = (2, 1, 1, 0)$

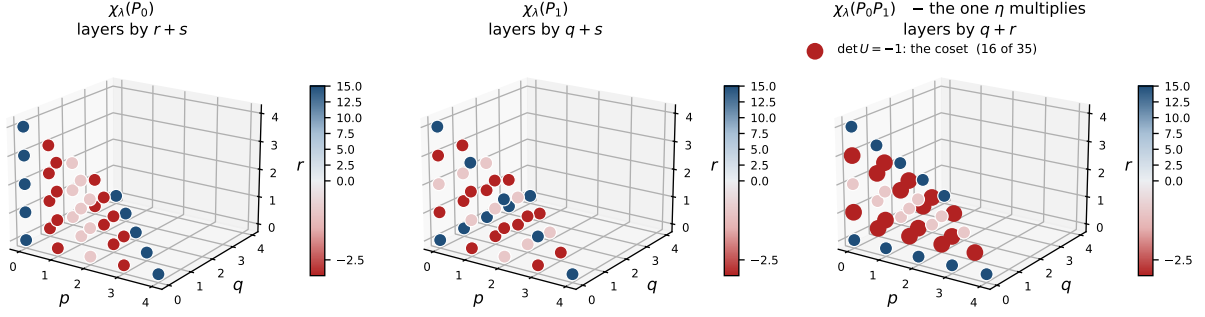


Figure 6: Proposition 3 drawn. A boundary condition of $SU(4)$ on $S^1/\mathbb{Z}_2$ is four non-negative integers $[p; q, r; s]$ summing to 4, so the set of them is a simplex with 35 points; each is placed at (p, q, r) , with s implied. The three panels colour the same simplex by the three characters of (17), evaluated on the gauge adjoint. The colour job is polarity, so the ramp is diverging with a neutral midpoint. **Each character layers the simplex in a different direction** — $\chi(P_0)$ depends on the block sizes only through $r + s$, $\chi(P_1)$ through $q + s$, and $\chi(P_0 P_1)$ through $q + r$; all three layerings are verified, not asserted, in `make_figure_simplex.py`. The third panel is the one that matters here: it is the character the boundary sign η multiplies, and the ringed points are the 16 boundary conditions with $\det U = -1$, the reflection coset — the only place in this picture where η can act at all.

fixed-point rotation matrix is a class invariant for *every* R ; we claim none of that. Proposition 3 is the projector count, a different use of the same three numbers, and the two meet in §7.1.

Turning on the boundary signs sends $(P_0, P_1) \mapsto (\eta_0 P_0, \eta_1 P_1)$, hence $\varepsilon_i \mapsto \eta_i \varepsilon_i$ in (17). Two consequences follow by inspection, and both are verified by sweep ($N \leq 6$, all $[p; q, r; s]$, three representations, 615 cases, 0 violations each):

1. $(\eta_0, \eta_1) \rightarrow (-\eta_0, -\eta_1)$ permutes the four multiplicities by $(\varepsilon_0, \varepsilon_1) \rightarrow (-\varepsilon_0, -\varepsilon_1)$. This is their own sentence below eq. (4.3), here as an identity: the overall flip is a redundancy.
2. The finite part of their potential uses only the pair sums $N^{(++)} + N^{(--)}$ and $N^{(+-)} + N^{(-+)}$ (their N_v in eq. (3.25), and eqs. (4.11)–(4.13)), and these are $\frac{1}{2}[\dim \pm \eta_0 \eta_1 \chi(P_0 P_1)]$: they depend on the boundary signs *only* through the product $\eta = \eta_0 \eta_1$, and they distinguish its two values exactly when $\chi(P_0 P_1) \neq 0$.

The individual sign η_0 survives only in their N_ξ term, and ξ is the divergent quantity they themselves declare incomparable between equivalence classes. Everything finite sees the product. This is why (15) is a definition and not an approximation.

So η is invisible to a multiplet precisely when the character at the winding element $U = P_0 P_1$ vanishes. For $SU(4)$, $\det U = (-1)^{q+r}$, so U lies in the reflection coset exactly when $q + r$ is odd — 16 of the boundary conditions, and U then has only two spectra, $\text{diag}(1, 1, 1, -1)$ and $\text{diag}(-1, -1, -1, 1)$, the conjugacy class of Part III’s parity matrix P_0 . Hence $\chi_\lambda(U) = s_\lambda(1, 1, 1, -1) = D_\lambda(1)$ and Theorem 1 reads, in their variables: $\eta_0 \eta_1$ is invisible to λ if and only if $\chi_\lambda(U) = 0$.

7.1 The Wilson line interpolates the trace conservation laws

The two readings meet, and the meeting point is the sharpest sentence we can make about our own object. A trace conservation law is a number: the character at a fixed-point rotation matrix. Our D_λ is a *function* of the Wilson line phase, and at $t = 1$ it is $\chi_\lambda(\text{diag}(1, 1, 1, -1))$ — the character at Part III’s parity matrix, that is, exactly a trace conservation law for the representation λ . So the two Schur specialisations of (5) interpolate the conserved traces: the invariants of [26, 27] are the endpoints, and the Wilson line phase is what runs between them. The theorem of §6 then

says that the boundary sign is invisible exactly where that interpolating function is identically zero — and the corollary below says that this is already decided at the endpoint.

A difference of regime that turns out not to exist

Their formulas are evaluated at vanishing Wilson line phase, where the sign multiplies the number $D_\lambda(1)$; ours lives at a dynamical phase, where it multiplies the polynomial $D_\lambda(t)$. A priori $D_\lambda(1) = 0$ is the weaker condition and should define a strictly larger class of blind multiplets. It does not.

Corollary 2. $D_\lambda(1) = 0 \iff D_\lambda \equiv 0$. *The boundary-condition sign is invisible at vanishing Wilson line phase if and only if it is invisible at every phase, and both conditions are $\mathcal{Z}$.*

Proof. Part IV’s closed form gives $\delta(m) = \zeta \cdot (\text{non-negative})$ with a single global sign ζ , so all coefficients of $D_\lambda = \sum_m \delta(m)t^m$ carry the same sign; a sum of terms of one sign vanishes only if every term vanishes. $\square$

In the language of §7.1: *a single conserved trace controls the whole interpolating function*, and it does so because Part IV’s closed form gives the coefficients one sign — so this corollary carries that identity’s standing, and no more.

[Verified over the 455 irreducibles in range: 0 have coefficients of both signs; $D_\lambda(1) = 0$ on 47 and $D_\lambda \equiv 0$ on the same 47, with 0 in either class and not the other.]

The practical consequence is that the theorem can be quoted to a five-dimensional reader and a six-dimensional one without changing a word.

What is theirs

Stated plainly, because the neighbourhood is theirs: the classification $[p; q, r; s]$, the equivalence relation $[p; q, r; s] \sim [p - 1; q + 1, r + 1; s - 1]$, the count $(N + 1)^2$, the diagonal-representative theorem, the generic one-loop potential at vanishing Wilson line phase, and the redundancy $(\eta_0, \eta_1) \sim (-\eta_0, -\eta_1)$ are all in [4]. So is a blindness of the potential: they observe that their eq. (4.13) “is a function of $q + r$ only” and “does not select unique $[p; q, r; s]$ ”, and close the paper saying the arbitrariness problem is unsolved because “there remains a degeneracy among the theories of the lowest energy density”. That degeneracy is on the side of the boundary-condition *labels*. Ours is on the side of the matter *content* — which multiplets cannot see η — and the two are complementary, in the same spirit, and not the same statement. What is added here is Proposition 3 for general representations, where they tabulate three by hand, and the closed-form identification of the blind class.

When a model has an index sector at all, and how often

The root of §1 is a test, and it is a test on the *embedding* rather than on the matter: a coset sector exists if and only if $\det P_0 \neq \det P_1$. In the $[p; q, r; s]$ labels that is $\det U = (-1)^{q+r}$, and the parity of $q + r$ is a class invariant — the equivalence move sends $q + r \mapsto q + r + 2$. So the question can be counted, not just posed:

N	classes $(N + 1)^2$	with an index sector	without
2	9	4	5
3	16	8	8
4	25	12	13
5	36	18	18
$\vdots$			
10	121	60	61

The pattern is exact, and it is a theorem rather than a sweep. The move leaves $(p + q, q + s)$ — that is, $(\#\{P_0 = +1\}, \#\{P_1 = -1\})$ — untouched, and that pair is a *complete* invariant: two block patterns with the same N and the same pair are joined by a chain of moves, because the pair itself leaves room for each step. So the classes of rank N are exactly the pairs (u, v) with $u, v \leq N$, which is [4]’s $(N + 1)^2$ recovered, and $q + r \equiv u + v + N \pmod{2}$, which is what makes “has a coset sector” a property of the class.

Proposition 4 (the census). *Of the $(N + 1)^2$ equivalence classes of boundary conditions of $SU(N)$ on $S^1/\mathbb{Z}_2$, exactly $\lfloor (N + 1)^2/2 \rfloor$ admit a reflection-coset sector, and in the other $\lceil (N + 1)^2/2 \rceil$ the boundary sign is invisible to every multiplet.*

Both halves are machine-checked in Lean 4 [29] (GHUCosetCensus.lean, sorry-free, the same three axioms as Table 1): `label_move` and `eqv_of_label` for the complete invariant, `coset_parity` for the class function, and `card_cosetClasses` for the count, for every N and not for the ten in the table. The controls are in the file and they can fail: the equally natural label $(p + q, q + r)$ is *not* move-invariant, the other parity class has the other cardinal, and a brute-force enumeration of every $[p; q, r; s]$ of rank three returns exactly the grid the theorem predicts.

This census is an $S^1/\mathbb{Z}_2$ statement and stays one. It is legitimate there because every equivalence class has a diagonal representative — that is [4]’s Appendix A, and without it the count would be over representatives rather than classes. The corresponding classification on the two-dimensional orbifolds, including the $T^2/\mathbb{Z}_2$ of this model, is Kawamura and Miura’s [25], and there the diagonal assumption is *not* free: they exhibit boundary conditions that cannot be simultaneously diagonalised at all. We therefore do not carry the count across, and §10 says so.

This closes the invisibility question into two disjoint cases, and only the second is ours:

*The boundary-condition sign η is invisible either because the geometry has no coset sector — $\det P_0 = \det P_1$, roughly half of all classes, and then it is invisible to **every** multiplet — or because the matter is parity-balanced, which is Theorem 1 and Part IV’s class $\mathcal{Z}$.*

The first case is a one-line check a model builder can run before anything else; the second needs the closed form. Stating both is what makes the answer complete rather than merely true.

Every section so far has been about what a boundary condition is. The variable that decides which part of one is physical has been present in every formula and has never once been the subject; §8 makes it one.

8 The Wilson line, and what it can and cannot absorb

The Wilson line has appeared in every section as a variable and has not once been the subject. It is the connective tissue of the whole construction, and the pieces above say five distinct things about it.

It is the Higgs. Not an analogue: the physical degrees of freedom are the gauge-invariant phases of $W = P \exp(ig \oint A_y dy)$. At tree level the potential is flat along them — they parametrise degenerate vacua — and the degeneracy is lifted only at one loop. That is the Hosotani mechanism [6], and it is the entire electroweak sector of this model.

It is the continuous half of the holonomy. The winding amplitude is the character at $\exp(i\pi\theta T) U_6^{k_2}$: a *continuous* Wilson line times a *discrete* orbifold twist. The discrete factor takes only two values in $O(4)$, and that is the $\mathbb{Z}_2$ of §3; the continuous factor is t , and it is what makes each half a *function* rather than a number. Everything Part IV established is a statement about that function, uniform in it.

It absorbs exactly the part of a boundary condition that is not physical. Two boundary conditions in one equivalence class are related by a transformation that *changes* them, and the equivalence holds because the Wilson line shifts to compensate — point (v) of the Hosotani

mechanism as [4] state it. That is why the trace conservation laws of [26, 27] are conserved: at a fixed point the transformation is conjugation by a constant, which the Wilson line cannot reach. Measured on the classification:

datum	under the class move	absorbed by
$\text{Tr } P_0, \text{Tr } P_1$	invariant	nothing — they <i>are</i> the class
$q + r$	$\mapsto q + r + 2$	the Wilson line
parity of $q + r$, i.e. $\det U$	invariant	nothing — it decides whether a coset sector exists

[all three rows are proved for every N , by Proposition 4]. *The class invariants are precisely what the Wilson line cannot absorb.*

It interpolates the conserved traces. Theirs are numbers at fixed points; ours is a function, and $D_\lambda(1)$ is one of their invariants (§7.1). Corollary 2 then says something sharp about the interpolation: for this object the Wilson line carries *no* information about vanishing — the endpoint already decides it — while carrying all of it about magnitude.

And it is the obstruction between a closed-form potential and a closed-form Higgs mass. $V(\alpha)$ is closed-form for arbitrary content (§2); the only transcendental step left is minimising over α . That is where this paper stops, and the reason is worth recording.

A draft of this section reported a measurement here — that at the vacuum the coset half carries the whole Higgs curvature — and it did not survive its own instrument check: the assembly which produced it missed AHMN’s published mass matrix, and on the corrected one the pattern is gone. The full record, and the corrected instrument, open Part VI. What survives of this section is its structure, not that measurement.

What survives is the negative that motivated the search and is unaffected by it: the Dynkin index does not fix the Higgs curvature at the vacuum, because it is the $\alpha \rightarrow 0$ limit and the Wilson line has moved the system elsewhere. *Which half carries the mass is open, and it is not this paper’s question* — see §10.

One distinction not to blur, since the two are easy to conflate and the literature warns against it: α is continuous, gauge-invariant and *is* the Higgs; η is a discrete boundary phase. They enter differently — α as the argument of both halves, η as a multiplier on the coset half alone — and that difference is the reason §6 has anything to say.

That is the structure, end to end. What it is for — and what it says to a problem the field has had open for twenty-two years — is the next section, and the last one that claims anything.

9 The reading

The declared open problem of this field — four groups over twenty-two years — is that no known mechanism selects among equivalence classes of boundary conditions. Hosotani named it the arbitrariness problem [12]; [4] attacked it by comparing vacuum energy densities and reported that a degeneracy survives; [5] is still working the same seam.

Theorem 1 speaks to it from the other side. It does not select a boundary condition. It says exactly *which multiplets cannot help you select one*, because for them the choice is invisible in principle — and it identifies that set with a criterion Part IV already classified in closed form, with no hypothesis on the shape of the partition, and which Proposition 1 counts. A model builder choosing bulk matter to make a boundary condition dynamically preferred now has a test that costs nothing: if the content lies in $\mathcal{Z}$, the coset half is empty and the content is inert with respect to η , no matter how the rest of the model is arranged.

What changes for someone building one of these models

The rest of this section is the practical content, stated as four consequences with the limit of each attached, because an implications section is where a paper overclaims if it is allowed to.

1. A selection rule on bulk matter, applicable by inspection. Given a Young diagram, Proposition 1 decides in closed form whether that multiplet lies in $\mathcal{Z}$. If it does, the multiplet contributes *nothing* to the coset half: it cannot help select a boundary condition, and it enters the breaking criterion (13) through its Dynkin index alone. That is a constraint one can apply before computing anything. *Limit:* it is per multiplet — a content is a multiset, and cancellations between multiplets are covered only where the effective signs agree (Proposition 2); the frustrated case is not.

2. The potential of an arbitrary multiplet, without a Kaluza–Klein spectrum. The published treatments obtain the mode list per representation — [24] for the adjoint, fundamental and (anti)symmetric tensors, [3] and [5] by hand from the spectrum. Here it is read off λ for *any* irreducible, through the same universal G (§2). The practical consequence is arithmetic: a scan over matter content stops being limited to the handful of representations whose spectra somebody has written out. *Limit:* a closed form for the potential is not a closed form for the Higgs mass, which needs the minimum.

3. A one-line test on the embedding, before any matter is chosen. A coset sector exists at all only when $\det P_0 \neq \det P_1$, and $\lfloor (N+1)^2/2 \rfloor$ of the $(N+1)^2$ equivalence classes are of that kind (§7). In the other half the boundary-condition sign is invisible to *every* multiplet, for reasons that have nothing to do with the matter. The count is Proposition 4 and holds for every N . *Limit:* it is an $S^1/\mathbb{Z}_2$ count, and diagonal boundary conditions only.

4. The two halves answer different questions, and that splits the next problem cleanly. The even-winding half determines the matter up to conjugation; the odd-winding half carries the boundary sign. For a content rather than a multiplet this matters, because the coefficients of Σ_λ are non-negative (§2) and therefore a sum of them *cannot cancel*: two contents with the same even half are two decompositions of one non-negative function, an additive question with no cancellation in it. The odd half is where genuine cancellation between multiplets lives — and by Proposition 2 it needs *sign frustration* to live there at all: with one effective sign $\eta_i \zeta_i$ throughout, the odd half cannot cancel either, and the content is blind only if every multiplet in it is. Whoever attacks the degeneracy of contents should attack those two halves separately, and inside the odd one, only the frustrated case; they are not the same problem. *Limit:* the separation statement for Σ_λ is verified on 969 multiplets and not proved (§10).

5. The weight in the breaking criterion is geometry, not a parameter. $\rho = 7.5115$ is fixed by the fall-off of the winding sum, hence by the dimension of the compactification, because the shortest winding that contributes to the curvature is odd (§5). It is not available for tuning — and it also says how fragile a verdict is: the base **60** of this model sits at a critical $\rho = 6.4545$ against an actual 7.5115, so a 14.1% shift reverses it. Verdicts within that margin should be quoted with it.

Part III exported a question about symmetric functions. The answer comes back as a statement about which part of the matter content is observable in principle: *the vanishing class is where the boundary condition stops mattering.*

Everything claimed here is negative — what cannot be seen, and therefore cannot be wrong. Running the same machinery forwards, to predict a Higgs spectrum rather than to rule one out, is a different kind of claim and a more fragile one. It is the subject of the sequel and is deliberately not in this paper. What is owed here are the limits.

10 Scope, and what is open

Proved. Relation (16), on any inversion-stable alphabet and for any number of reciprocal pairs. The identification of its degenerate case with branch (b) of Part IV. Proposition 3. Proposition 1,

machine-checked, and with it the vanishing criterion itself — Part III’s theorem in Dynkin labels, whose dictionary with the partition form §6.1 certifies. Proposition 4, machine-checked, and with it the completeness of $(\#\{P_0=+1\}, \#\{P_1=-1\})$ as a class invariant: the census holds for every N , where the paper’s table only sweeps ten.

Proved on an identity that is itself verified and not proved. Corollary 2, and every one-sign argument, needs Part IV’s closed form to give D_λ coefficients of a single sign, and Part IV states that identity as an Observation. What does *not* depend on it is the criterion the identity computes, so $\mathcal{Z}$, its count and its Lean certificate stand regardless; only the sentence that $\mathcal{Z}$ is the *zero locus* of D_λ would have to be re-read.

Verified, not proved — and now reduced to one function and one residual case. Observation 1: that (16) generates *all* collisions is verified with 0 exceptions among 910 pairs, and unproved. Three measurements sharpen the problem considerably, and they are worth stating because they turn a search into a question.

First, the pair is not needed: Σ_λ *alone separates*. Over the 969 multiplets with $\lambda_1 \leq 16$ its fibres have sizes $\{1 : 81, 2 : 444\}$, the 81 singletons are exactly the 81 self-conjugate multiplets, and *no* fibre fails to be the conjugation orbit $\{\lambda, \lambda^*\}$. So the question is about one symmetric function, not two.

Second, two of the three parameters come out in closed form, and both are conjugation-invariant as they must be: the top exponent of Σ_λ is λ_1 , and its top $SU(2)$ coefficient is $\lambda_2 - \lambda_3 + 1$ [both 0 failures on all 455 multiplets with $\lambda_1 \leq 12$]. Since $\lambda^* = (\lambda_1, \lambda_1 - \lambda_3, \lambda_1 - \lambda_2, 0)$ leaves λ_1 and $\lambda_2 - \lambda_3$ fixed, neither could have distinguished the orbit, and neither does.

Third, what is left is $\lambda_2 + \lambda_3$ modulo the reflection $s \mapsto 2\lambda_1 - s$ that conjugation induces — and the next two coefficients do *not* determine it. There are 28 collisions, and they are all of one shape: the family $\lambda = (\lambda_1, k, k, 0)$ has the same first three $SU(2)$ coefficients for every k . The full coefficient sequence does separate them; a proof has to reach past the top of the list, and the residual case is now named.

Per multiplet. A physical content is a multiset of irreducibles with multiplicities and the observable is the *sum*, so distinct contents may still collide by cancellation even when no two multiplets do — but only with both effective signs present, by Proposition 2. How large that frustrated degeneracy is, and what generates it, is open and is the next paper. The tools for it are exact ones — integer linear programming over the anomaly and tadpole constraints, quotiented by the invisible group identified here — and the moment tower $M_{2r} = \sum_m \delta(m)(m^2)^r$ is a signed sum of geometric sequences, so its Hankel matrix has rank equal to the number of distinct winding charges and a generalised eigenvalue problem returns those charges exactly, with integer eigenvalues [hankel_probe.py].

Not claimed. That the potential distinguishes R from $\bar{R}$: it does not, see §6, and the literature says so first. That anything here is about nature: every statement is about a model, on $T^2/\mathbb{Z}_2$ with $U_5 = 1$, at one loop.

Diagonal boundary conditions only, and we did not know this until we read for it. Every statement above assumes the orbifold parities can be simultaneously diagonalised, which on $S^1/\mathbb{Z}_2$ is no restriction — Haba, Hosotani and Kawamura prove every equivalence class has a diagonal representative [4]. On $T^2/\mathbb{Z}_2$, the geometry of this model, it *is* a restriction: Kawamura and Miura [25] exhibit boundary conditions for $SU(N)$ ($N \geq 3$) that cannot be brought to diagonal form by any global unitary together with a local gauge transformation, and note that a multiplet’s components then fail to be simultaneous eigenstates of the $\mathbb{Z}_2$ parities. The two alphabets of (3) presuppose a diagonal winding element. Nothing here applies to the off-diagonal classes, and we make no claim about them.

Deliberately not here: everything that runs the machinery forwards. Which half carries the Higgs mass at the vacuum, what a given matter content predicts for $\alpha_{\min}$ and κ_V , and how large the degeneracy becomes once contents carry multiplicities — all of that is an open search, and a closed theorem and an open search do not belong in one paper. They are Part VI. This paper is the closed half, and every statement in it is either proved or exhaustively verified

in a stated range.

And the standing lesson, since it has now cost us twice. A measured pattern is worth nothing until the instrument that measured it reproduces something published. Ours reproduced AHMN’s *vacuum* approximately and their *mass matrix* not at all, and the difference was a twist recorded in our own archived output and never applied.

A neighbouring literature, and what it does and does not say about this. C^* boundary conditions [28] make fields periodic *up to charge conjugation*, so that one trip round the torus applies an automorphism that is not inner. Three features there have the shape of what is above: a $\mathbb{Z}_2$ survives the twist (electric charge is not conserved, but $(-1)^Q$ is, their eq. (2.26)); the effect comes entirely from a particle that winds at least once; and the boundary data is a set of discrete signs modulo one overall redundancy (their eq. (6.7)). We take the framing and claim nothing from it: their $\mathbb{Z}_2$ is a *conservation law on states* — charged states cannot mix with the vacuum — whereas the winding parity here is a *grading of a sum*, an index of summation rather than a quantum number. The shape is shared; the statement is not, and we do not borrow it.

Untouched. The viability of the (3,60) content under the boundary-mass filter, which is a question of Part I and is not addressed by anything above. And the cosmological-constant question is not ours to touch: the winding sum excludes $k = 0$, and the $k = 0$ term is exactly the α -independent piece that would be it.

11 Ledger

Item by item, as in Parts III and IV. Non-negotiable, and rebuilt whenever the paper moves.

Inherited	Source
The Hosotani mechanism; Wilson-line phases as physical degrees of freedom	[6]
The orbifold construction, chiral fermions and the split multiplet	[9, 10], with [11] for the supersymmetry-breaking twist
Boundary-condition equivalence classes on $S^1/\mathbb{Z}_2$; $[p; q, r; s]$; the count $(N + 1)^2$; the diagonal-representative theorem; the redundancy $(\eta_0, \eta_1) \sim (-\eta_0, -\eta_1)$; the arbitrariness problem	[4], [12]
The classification on the two-dimensional orbifolds, and the existence of boundary conditions that cannot be simultaneously diagonalised	[25]
Trace conservation laws: the character at a fixed-point rotation matrix is an equivalence-class invariant, and the classification built on them	[26, 27]
A general one-loop potential per representation class, and the method that extends it	[24]; the closest two-phase computation, [8]
The published one-loop potential of the 6D $SU(4)$ model and its coefficients $\{A+B(-1)^{k_2}\}$	[3]
That charge conjugation is invisible in this class of models	[5]
The centre-charge grading of the Wilson-line potential	[13], [14], [15, 16]
That a root-of-unity specialisation makes a character vanish off the t -core and factor over the t -quotient	Littlewood–Richardson, and [21, 22, 23] — none of them our alphabet
The twining reading of a character on a non-identity component	[19, 20], via Part IV
A non-inner boundary twist, and discrete boundary signs modulo one redundancy, as standard practice	[28]

Ledger, continued	Source, evidence, or reason
The closed form for $s_\lambda(1, -1, t, t^{-1})$, its vanishing criterion, and ζ as a parity index	Part IV [2], which states the closed form as <i>verified</i> , <i>not proved</i>
The mode labels, $U_6 = P_2 P_0^{-1}$, the notch and the decoupling corollary	Part III [1]
Ours	Where, and on what evidence
Winding parity is the $O(4)$ component label, so the potential is $(\Sigma_\lambda, D_\lambda)$; and $\det P$ is pinned by $P^2 = 1$, not chosen	§1–2
Equation (7): the two halves are one operator traced twice, $\text{Tr}(g)$ and $\text{Tr}(\sigma g)$ — a dimension and an index	§2, 287 checks, 0 failures
Both halves as stacks of one universal box; the identity half as a positive superposition, with a summation range that is not a choice	§2, 286/286, 2197/2197
Why one half factors and the other cannot: the frozen letters are roots of unity, so every zero of D lies on the unit circle and essentially none of Σ ’s do — and the <i>control</i> , the repeated-letter blocks, is ours	§2, Figure 3
The dictionary $A = \frac{1}{2}(\Sigma + D)$, $B = \frac{1}{2}(\Sigma - D)$, and the published coefficients of three separate papers reproduced from representation theory	§4; [3] twelve numbers, [24] four printed objects, [8] two of three representations
$\rho = 7.5115$ is geometry and not a coupling: the shortest winding that contributes to the curvature is odd	§5, control at 0.3443
Theorem 1: the classes of (λ, η) , and η blind exactly on $\mathcal{Z}$	§6, 910 $\rightarrow$ 470, 0 unexplained
Proposition 1: $\mathcal{Z}$ characterised and counted, $\lfloor (k+1)^2/2 \rfloor$ and never λ_1 even	§6.1, Lean 4; 0 disagreements on 2925
Corollary 1: $\mathcal{Z}$ is level-by-level parity balance	§2, 47 = 47
Proposition 3: a boundary condition sees a representation through three characters	§7, 1470 cases, 0 mismatches
Corollary 2: the two regimes are one condition	§7, 47 = 47
Proposition 2: collective blindness requires sign frustration	§6, one line from Part IV’s one-sign property, and with its standing
Proposition 4, the census: $\lfloor (N+1)^2/2 \rfloor$ of the $(N+1)^2$ classes admit a coset sector, splitting invisibility into a geometric case and a matter case	§7, Lean 4; every N , not the ten in the table
That Σ_λ alone separates the multiplet up to conjugation, with λ_1 and $\lambda_2 - \lambda_3$ recovered in closed form — <i>verified</i> , <i>not proved</i>	§10, 969 multiplets, 0 exceptions
Retracted, and left visible	Why
“The coset half is the only observable that sees charge conjugation”	false, and [5] say so in print. §6
“At the vacuum the coset half carries the whole Higgs curvature”	an artefact of an assembly that missed [3]’s mass matrix. §10, and Part VI

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